ISAAC SIM/LAB BENCHMARKS

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PLATFORM OVERVIEW

01

DGX SPARK

GB10 Blackwell
ARM Cortex-X925 (20)
125 GB unified

02

SPIKE-1

B200 Blackwell
Intel Xeon Platinum 8570 (224)
183 GB discrete

03

SURF RESEARCH CLOUD

A10 Ampere
Intel Xeon Gold 6342 (11)
24 GB discrete

04

SNELLIUS

A100 / H100
Intel Xeon Platinum 8360Y (72)/ AMD EPYC 9334 (64)
40 / 96 GB

OUR PLATFORMS

DGX SPARK

- Direct access — keyboard, monitor, or SSH
- No job scheduler, no container needed
- Isaac Sim runs from local source build

SPIKE-1 (B200)

- Access via Run:AI job scheduler
- Runs containerised workloads (Harbor registry)
- GUI via TurboVNC, headless via SSH port-forward

SURF PLATFORMS

SURF RESEARCH CLOUD

- Browser-based GUI via Guacamole (no SSH required)
- Workspace provisioning via SURF portal
- Isaac Sim runs from standard NGC container

SNELLIUS

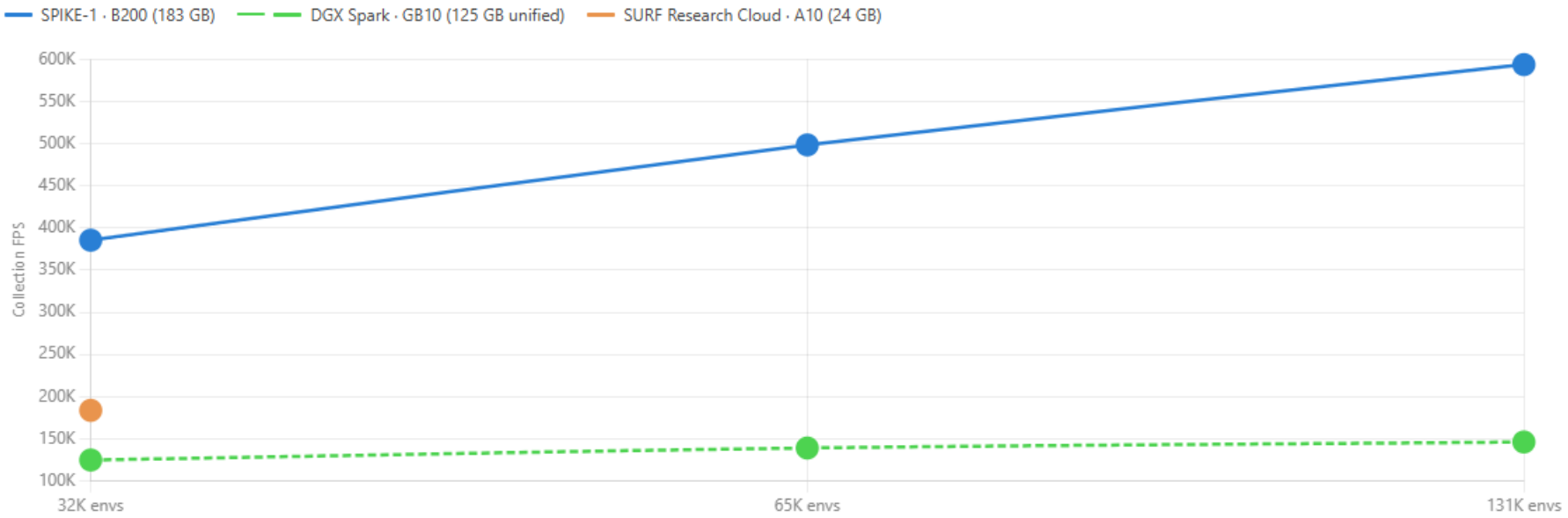
- Access via SLURM (sbatch) job scheduler
- Runs via Apptainer (Singularity) containers
- Requires active SURF SRAM account

Results!

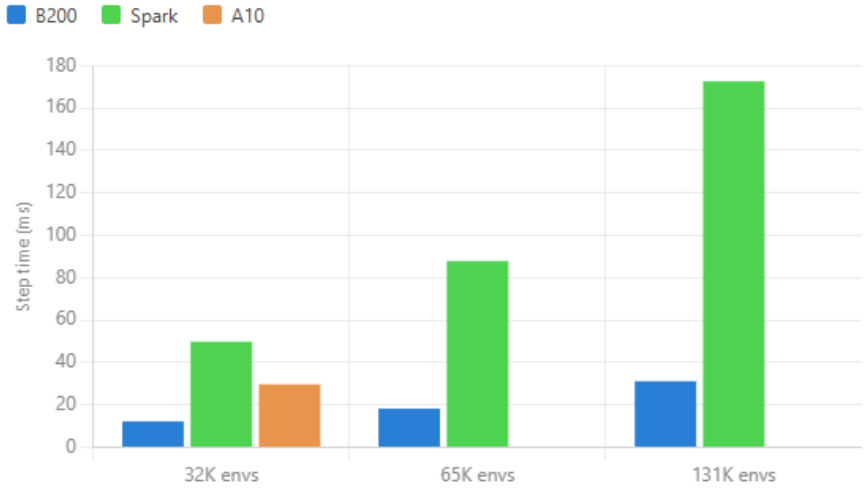
From a total of 14 different benchmarks, across all of the mentioned platforms!

1 — HEADLESS RL COLLECTION FPS VS ENVIRONMENT COUNT

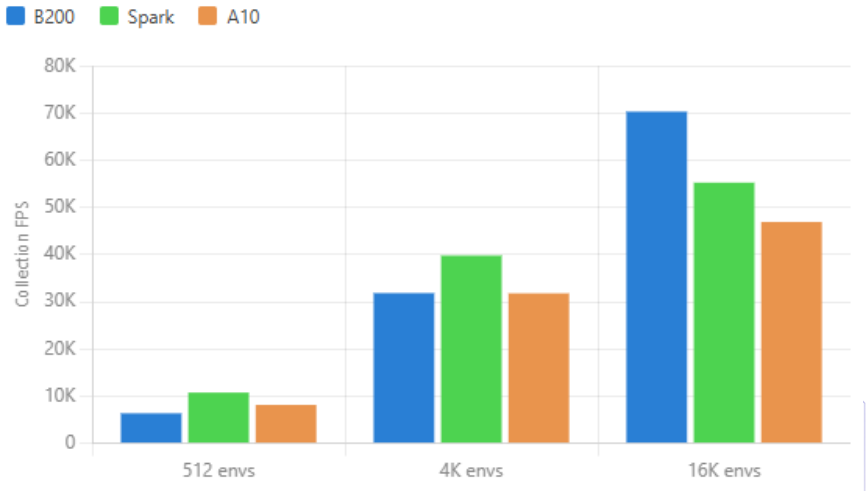
· ANYMALD LOCOMOTION · PPO



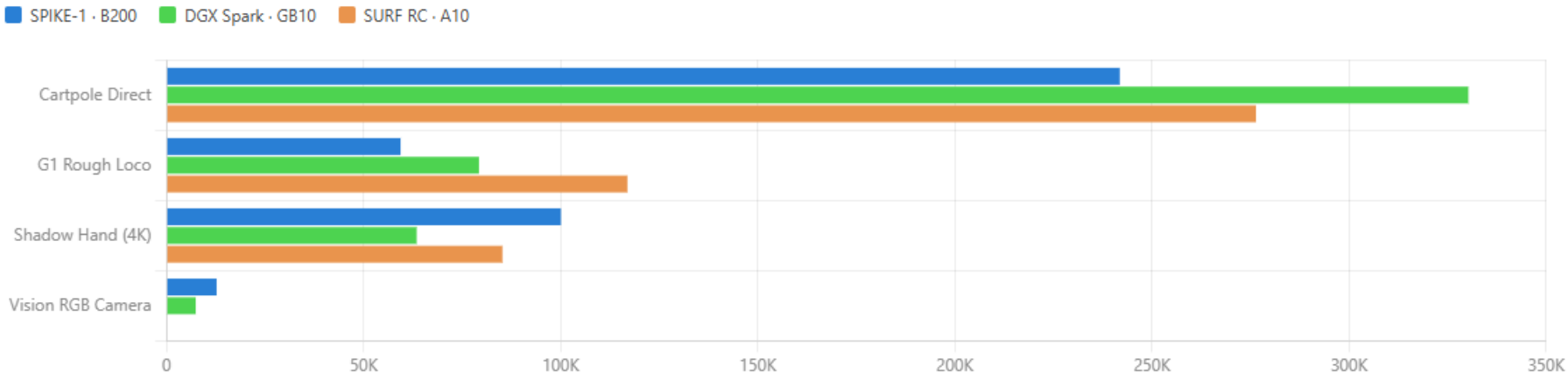
2 — PHYSICS STEP TIME (MS) AT SCALE
• LOWER IS BETTER



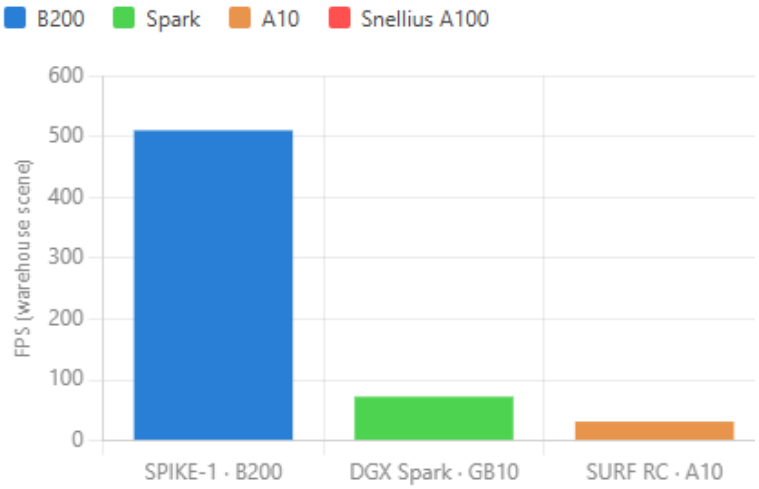
3 — GUI MODE COLLECTION FPS
• RENDERING OVERHEAD INCLUDED



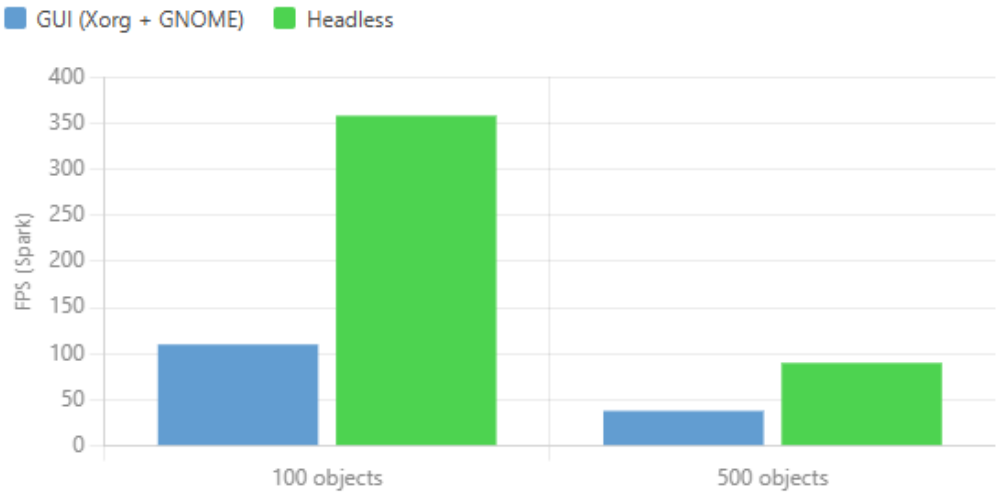
4 — ISAAC LAB OFFICIAL BENCHMARKS · TRAINING FPS · 4K ENVS



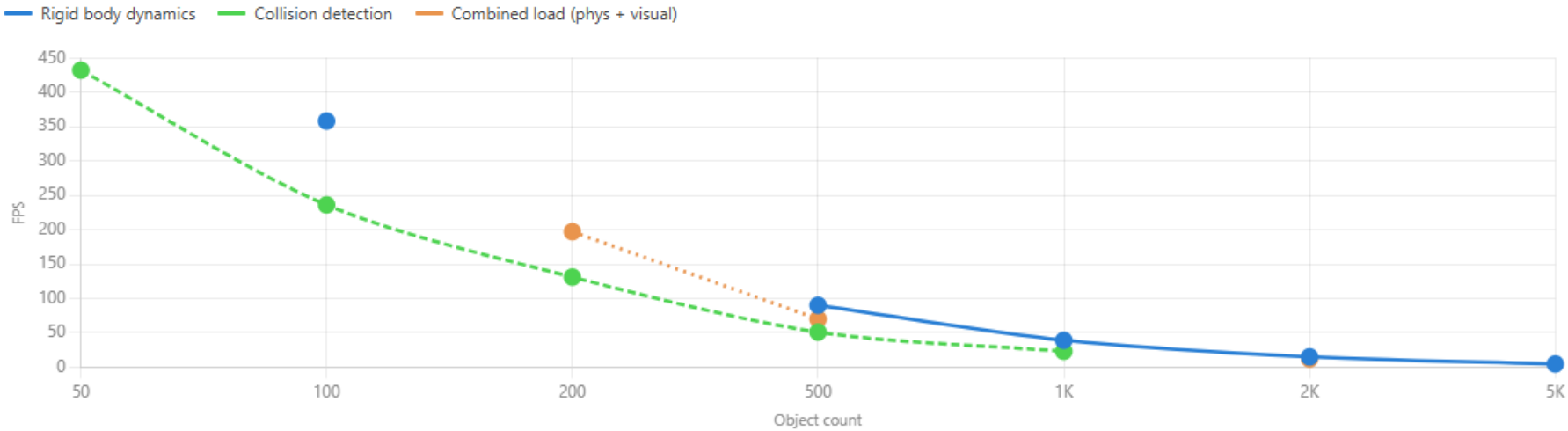
5 — SCENE RENDERING FPS • WAREHOUSE • RT2 MODE



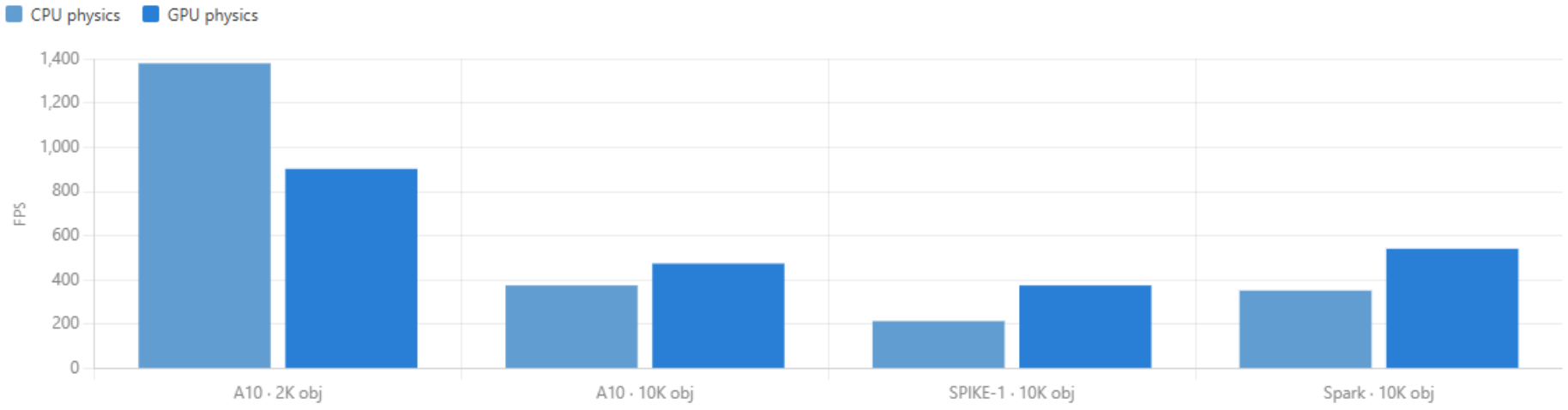
6 — GUI VS HEADLESS OVERHEAD DGX SPARK



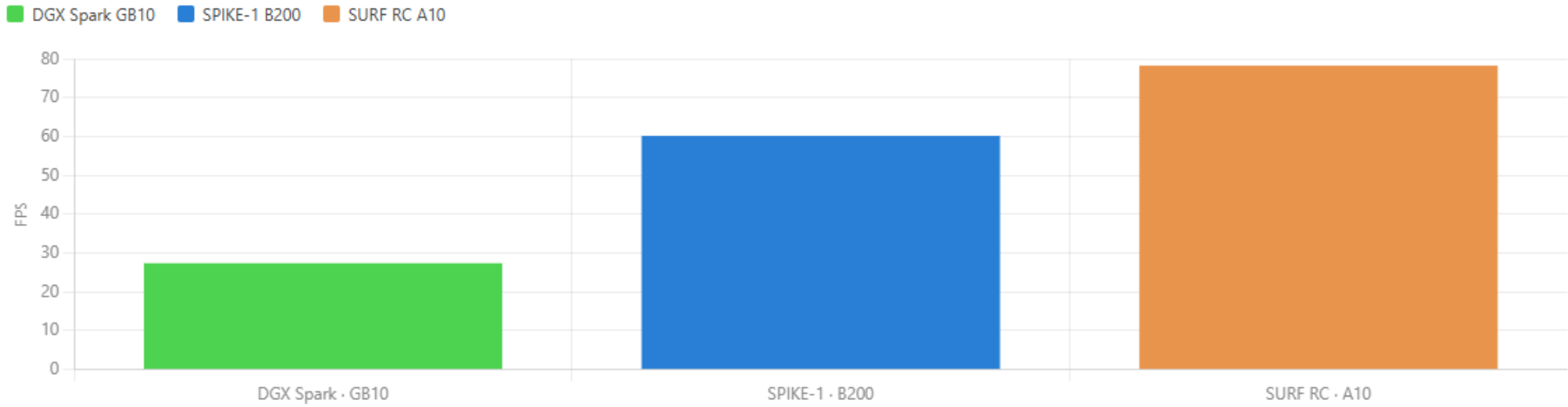
7 — SPARK RIGID BODY, COLLISION & COMBINED LOAD FPS VS OBJECT COUNT



8 — PHYSX GPU VS CPU CROSSOVER LOW-LEVEL RIGID BODY FPS



9 — SCC BENCHMARK V2 · 10× FRANKA ROBOTS · PHYSICS-ONLY · HEADLESS



Now what?

The Conclusions and Next Steps ofcourse!

CONCLUSION

It depends entirely on what you're doing.

- Designing and developing robots → **DGX Spark**
- Training RL policies at scale → **SPIKE-1**
- Onboarding new researchers or occasional use → **SURF Research Cloud**
- Vision-based RL (camera observations) → **SPIKE-1 only**
- Pure compute / LLMs / non-Isaac workloads → **Snellius**



NEXT STEPS

Try **rescale** as an alternative to SURF Research Cloud

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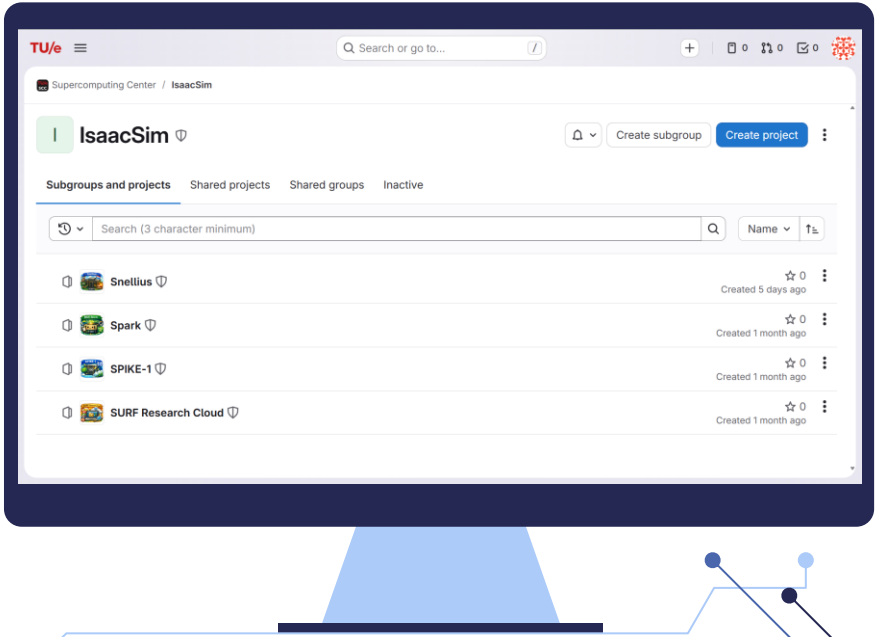
Wait till SRC ships the new RTX 6000 Pro‘s :)

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Test with multiple of the same GPUs

THANK YOU

All the documentation and data can be found on GitLab:
https://gitlab.tue.nl/hpclab/isaacsim



RAW DATA

ISAAC LAB RL · ANYMALD LOCOMOTION (PPO · SEED=42 · 10 ITER) · HEADLESS MODE

PLATFORM	GPU	ENVS	COLLECTION FPS	TOTAL FPS	PHYSICS STEP (MS)	STATUS
SURF RC	A10	32,768	183,741	167,255	29.74	OK
SURF RC	A10	65,536	—	—	—	CUDA OOM
Spark	GB10	32,768	124,941	111,539	49.83	OK
Spark	GB10	65,536	139,156	122,718	87.89	OK
Spark	GB10	131,072	146,302	128,081	172.54	OK
Spark	GB10	262,144	—	—	—	OOM killed
SPIKE-1	B200	32,768	385,420	365,904	12.30	OK
SPIKE-1	B200	65,536	498,333	476,324	18.30	OK
SPIKE-1	B200	131,072	593,513	569,855	31.20	OK

ISAAC LAB RL · ANYMALD LOCOMOTION · GUI MODE (RENDERING OVERHEAD INCLUDED)

PLATFORM	GPU	ENVS	COLLECTION FPS	TOTAL FPS	PHYSICS STEP (MS)	STATUS
Spark	GB10	512	10,755	10,173	3.45	OK
Spark	GB10	4,096	39,810	37,724	9.09	OK
Spark	GB10	16,384	55,265	52,234	31.21	OK
Spark	GB10	32,768	—	—	—	CUDA OOM
SPIKE-1	B200	512	6,364	6,117	3.84	OK
SPIKE-1	B200	4,096	31,854	31,021	6.99	OK
SPIKE-1	B200	16,384	70,322	69,215	14.65	OK
SPIKE-1	B200	32,768	—	—	—	Crashed
SURF RC	A10	512	8,146	7,626	3.01	OK
SURF RC	A10	4,096	31,798	30,737	6.75	OK
SURF RC	A10	8,192	41,228	40,020	11.86	OK
SURF RC	A10	16,384	46,931	45,624	22.79	OK
SURF RC	A10	32,768	—	—	—	CUDA assert

ISAAC LAB OFFICIAL BENCHMARKS · TRAINING FPS · 4,096 ENVS (UNLESS NOTED)

TASK	ENVS	SPARK GB10	SPIKE-1 B200	SURF RC A10
Cartpole Direct	4,096	330,271	241,811	276,391
G1 Rough Locomotion	4,096	79,235	59,351	116,908
Shadow Hand Cube Repose	4,096	63,436	100,040	85,195
Shadow Hand Cube Repose	8,192	—	107,314	85,195
Cartpole RGB Camera (vision)	1,024	7,386	12,684	OOM

SCENE RENDERING · WAREHOUSE · RT2 MODE

PLATFORM	GPU	FPS	RT FACTOR	GPU MEM
SPIKE-1	B200	510.2	3.63×	1.77 GB
Spark	GB10	72.4	1.16×	2.56 GB
SURF RC	A10	31.4	0.52×	3.46 GB
Snellius	A100	—	—	No RT cores

SPARK · RAY-TRACING · OBJECT COUNT & RESOLUTION SWEEP

SCENE / RESOLUTION	FPS	FRAME TIME (MS)
50 objects	39.7	25.2
100 objects	32.3	30.9
200 objects	32.2	31.1
500 objects	29.8	33.6
1920×1080	32.0	31.2
2560×1440	31.8	31.5
3840×2160	32.0	31.3

SPARK · RIGID BODY PHYSICS (CPU BASELINE)

OBJECTS	FPS	RT FACTOR
100	358	5.97×
500	90	1.49×
1,000	39	0.65×
2,000	15	0.25×
5,000	4.6	0.08×

SPARK · COLLISION DETECTION

OBJECTS	FPS
50	431.6
100	236.0
200	131.2
500	51.1
1,000	23.1

SPARK · COMBINED LOAD (PHYSICS + VISUAL)

SCENE	FPS
200p + 300v	196.6
500p + 500v	69.8
500p + 1,000v	79.8
1,000p + 2,000v	31.3
2,000p + 3,000v	12.1

PHYSX GPU VS CPU CROSSOVER · LOW-LEVEL RIGID BODY (DIRECT SIMULATE() CALLS)

PLATFORM	GPU	OBJECTS	CPU FPS	GPU FPS	DIFFERENCE	WINNER
SURF RC	A10	2,000	1,380	903	−35%	CPU
SURF RC	A10	10,000	376	475	+26%	GPU
SPIKE-1	B200	10,000	215	376	+75%	GPU
Spark	GB10	10,000	353	542	+53%	GPU (unified)
Spark	GB10	10 robots	122.4	103.0	−16%	CPU

SCC HPC BENCHMARK V2 · 10× FRANKA ROBOTS · HEADLESS · PHYSICS-ONLY

METRIC	SPARK GB10	SPIKE-1 B200	SURF RC A10
Mean FPS	27.4	60.2	78.3
Real-time factor	0.46×	1.00×	1.31×
Physics step (ms)	33.5	15.0	11.4
GPU utilization	6.4%	18.8%	0.0%
GPU power (avg)	10.9 W	242 W	58 W

SPARK · SUSTAINED 60-SECOND LOAD · 2,000 RIGID BODIES

METRIC	VALUE
Average FPS	18.48
Min FPS	18.32
Max FPS	18.90
FPS variance	0.58
Stability	96.9%
Duration / steps	60.01 s / 1,109
Thermal throttling	None observed

SNELLIUS · SLURM JOB RESULTS · ISAAC SIM 5.1.0 · DRIVER 590.48.01 · APRIL 2026

JOB ID	GPU	BACKEND	STARTUP	FRAMES	OUTCOME
21917463	A100 (40 GB)	numpy (CPU)	102 s	0 frames	Toolchain incompatibility — benchmark exits clean with zero measurements
21917467	A100 (40 GB)	warp (GPU)	100 s	0 frames	PhysX GPU invoked (confirmed by memory error) — framework still collects 0 frames
21917468	H100 (96 GB)	numpy (CPU)	97 s	0 frames	Same toolchain issue — H100 hardware is fine
21917469	H100 (96 GB)	warp (GPU)	crash T+88s	0 frames	Fatal crash: libgpu.foundation.plugin.so — driver 590.x + H100 + Vulkan incompatibility